

SIMATS
ENGINEERING



SIMATS

Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0506- COMPUTER VISION
MODEL LAB PRACTICALS
SET 10

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Slot-D

1. Perform Perspective Transformation on the given image using python and Open CV.
2. Perform basic Image Handling and processing operations on the image is to read an image in python and detect the corners in the image using Harris Corner Detection function
3. Implement a Sobel algorithm using Open CV to filter the input image.
4. Design and implement a water marking technique to insert the watermark into the original effectively image using Open CV.

viva answered

100

Aim: To perform Perspective transformation on a given image using Python and opencv.

Procedure:

1. Read the image using `cv2.imread()`
2. Select four corner point in the original image.
3. Define four corresponding points in the output image.
4. Calculate the perspective transformation matrix using `cv2.getPerspectiveTransform()`.
5. Apply the transformation using `cv2.warpPerspective()`.
6. Display original image and output image.

Code:

```
import cv2
import numpy as np
```

```
img = cv2.imread("input.jpg")
```

```
Pts 1 = np.float32([[50, 50], [300, 50], [300, 300], [50, 300]]);
```

$Pts2 = np.float32([0, 0], [400, 0], [400, 400])$

$matrix = cv2.get_perspectiveTransform(Pts1, Pts2)$

$result = cv2.warpPerspective(img, matrix, [400, 400])$

$cv2.imshow("original", img)$

$cv2.imshow("perspective", result)$

$cv2.waitKey(0)$

$cv2.destroyAllWindows()$

Input:



original image



Output:

The original image is displayed with the perspective transformed image.

Result: Hence, the program Executed Successfully using Python and opencv.

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Aim: To perform basic Image handling and processing operations and detect corners using Harris corner Detection.

Procedure:

1. Read the image using opencv
2. convert image into grayscale
3. convert grayscale image into floating-point format
4. Apply the Harris Corner detector.
5. mark the detected corners on the original image
6. Display the result.

Code:

```
import cv2
import numpy as np
img = cv2.imread("input.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
gray = np.float32(gray)
Corners = cv2.Corners_Harris ( gray, 2, 3, 0.04)
img [Corners > 0.01 * Corners.max()] = [0, 0, 255]
cv2.imshow ("Harris Corners", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input:



Output:

The detected corners are highlighted in red on the image.

Cornel detection.

Result:

Hence, the program Executed Successfully.

③

Aim: To implement the sobel algorithm using opencv filtering and edge detection of an input image.

Procedure:

1. Read the input image.
2. Convert it into grayscale.
3. Apply the sobel operator in the x direction.
4. Apply the Sobel operator in the y direction.
5. Combine x and y gradient's.

Code:

```
import cv2
import numpy as np
img = cv2.imread('input.jpg', 0)
```

```
sobel-x = cv2.sobel(img, cv2.CV_64F, 1, 0, ksize=3)
sobel-y = cv2.sobel(img, cv2.CV_64F, 0, 1, ksize=3)
```

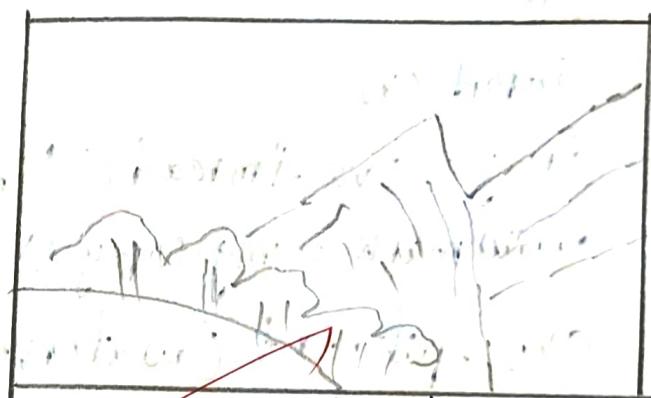
```
result = cv2.magnitude(sobel-x, sobel-y)
```

```
cv2.imshow("sobel filter", result)
```

```
cv2.waitKey()
```

```
cv2.destroyAllWindows()
```

Input:



Edge detection.

Output: The original image is displayed with sobel algorithm image.

Result: Hence, the program executed successfully using python and opencv.

(ii)

Aim: To design and implement a simple watermarking technique to insert a watermark into an original image.

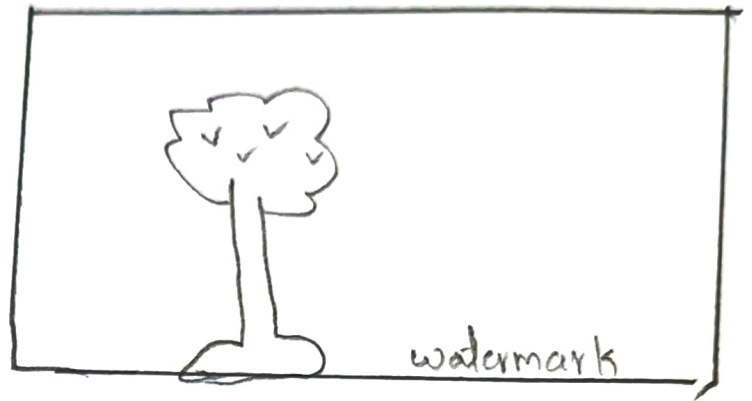
Procedures

1. Read the original image
2. create a watermark containing text
3. place the watermark at a selected position.
4. Blend the watermark with the image using `cv2.addWeighted()`.
5. Display and Save the watermarked image.

Code

```
import cv2
img = cv2.imread('input.jpg')
watermark = img.copy()
cv2.putText(watermark, "abcdxyz",
            (50, 100),
            cv2.FONT_HERSHEY_SIMPLEX,
            1, (255, 255, 255), 2)
result = cv2.addWeighted(img, 0.8, watermark, 0.2, 0)
cv2.imshow("watermark", result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

input:



output: The original image is displayed with watermarked image

Result: Hence, the program executed successfully using python and opencv.